

M13-C3 — H7b cautious `asm_intro(relto)` vs `asm_intro(sechk)`

H7b run / analysed / not a Bucket 3 claim

1 Status

H7b run / analysed. Bounded implementation-behaviour investigation of inherited ABALearn `asm_intro` under **cautious nd** on H0 and H3 `n30_seed42`. Markdown twin: `h7b_cautious_relto_vs_sechk.md`.

Lead (Q1–Q2): on mechanism-relevant child *c*, both arms **solved** but deltas differ — same `gen.pl` fork as H7a, under cautious semantics. Cautious `sechk` on *c* *retains* mechanism parents *b/d* (unlike brave H7a `sechk`) — bounded observation, not a claim. Comparator is locked `baseline_cautious` (H4/H4b), not `ecai2024`. Config `nd_cautious_sechk` is experimental. H0–H6 / H4 / H4b / H7a closed. **Not a Bucket 3 claim.**

2 Research question

Holding fixture, frozen sample, encoding, **cautious** semantics, nd folding, and all other learner options constant: how does changing `asm_intro(relto)` to `asm_intro(sechk)` affect what ABALearn learns and how search proceeds under cautious nd?

3 Main finding — contrast on child *c*

3.1 They diverge

Fixture 1 *c* and Fixture 2 *c*: both arms **solved**; deltas differ; inputs byte-identical within pairs.

3.2 Why

After failed post-fold entailment:

- **Relto:** relative-first; often K0 under cautious `gen3`; backtrack to another fold literal.
- **Sechk:** always mint provisional α ; `gen5`/stable extension; often still reaches *b/d*, but later / via deeper nests.

First fork:

```
relto:  looking for assumption relative to
sechk:  generating NEW assumption: alpha_1(A)
```

H7a’s brave first-BK latch does *not* describe these cautious-*c* sechk deltas.

3.3 What it looks like

Q1: same *a*-gate; sechk inserts α_2 nest then closes with `b_val_1` (relto closes with `b_val_0`).

Q2: both cite *a, b, d*; sechk thicker (5 vs 3 asms; ~899 vs ~257 lines).

4 Setup

Relto: `configs/baseline_cautious_config.pl`. Sechk: `configs/nd_cautious_sechk_config.pl` (experimental). Intervention: only `asm_intro`. Held fixed: cautious, nd, steps 10, any, all, `check_ic`, `post_folding_test_entailment(true)`. Paths: `.../<fixture>/{baseline_cautious,nd_cautious}`

Role discipline: only *c* is a desirable det. mechanism target; H0 *a* is a parent; H3 *a* is isolated; H3 *b* is role-peer of H0 *a*.

5 Outcome scaffold (secondary)

Cell	Relto	Sechk	Comparable?
Collider <i>c</i>	solved	solved	yes — deltas $\neq$ (Q1)
Collider <i>a, b</i>	no-sol	no-sol	cost only (Q3 uses <i>b</i>)
H3 <i>c</i>	solved	solved	yes — deltas $\neq$ (Q2)
H3 <i>a, b, d</i>	timeout	timeout	no — empty stdout (Q4–Q5)

6 Q1 — Fixture 1 *c*

```
% relto (~134 lines)
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
```

```
% sechk (~200 lines)
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- b_val_1(A).
```

Sechk is not “ignoring *b*”: extra assumption layer on the leading feature, then `b_val_1` rather than `b_val_0`. Vs H7a brave sechk on same *c*: brave dropped *b*; cautious retains it (bounded).

7 Q2 — Fixture 2 *c*

Both solve; both cite *a, b, d*. H4b relto lean (3 asms, 257 lines); sechk thicker (5 asms including $\alpha_{31}, \alpha_{38}, \alpha_{39}$; 899 lines). Same fork as Q1, amplified by the harder H3 table.

8 Q3 / Q4–Q5

Q3 Fixture 1 *b*: no delta either side; same token ladder 2...10; sechk stdout 13818 vs 5118 lines — failed-search cost only.

Q4–Q5 Fixture 2 *b* and *a*: timeout@300s both arms; stdout 0 bytes — non-comparison; not “matched search”.

9 Boundaries / non-claims

- Not a Bucket 3 claim.
- Not a ranking of relto vs sechk.
- **solved** $\neq$ mechanism recovery.
- Empty-log timeouts $\neq$ matching-search evidence.
- H7a brave latch does not carry over unchanged.
- H4 / H4b / H7a readings preserved.

10 Next

H7b documentation complete. Still **no Bucket 3 claim**.